

Primal Calculus

A Typed Reconstruction Calculus for Agreement, Defect, and Projection

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Corrected foundational paper – June 2026

Abstract

Primal Calculus is a typed reconstruction calculus for studying when multiple sectorwise reconstructions of common latent comparison data agree in a shared equality carrier. Its central object is the Aleph Operator, defined as the cyclic residual of a fixed-phase reconstruction tuple. The zero set of that residual is the exact agreement locus; finite residual sublevel sets are approximate agreement loci; the residual differential supplies a local regularity test and a normal defect form; and projection from the exact locus supplies tangential image geometry.

The algebraic and differential ingredients are standard: equalizers, cyclic coboundaries, constant-rank regularity, and pullback metrics. The contribution of the calculus is not the invention of those ingredients. It is the disciplined typed interface that keeps latent agreement, projected image behavior, defect stiffness, approximation, admissibility, and native certificate burden mathematically separate. This corrected version removes speculative application references, clarifies the role of the cyclic triple as a minimal model rather than a universal necessity, and adds a worked diagnostic example showing how the calculus prevents projection singularities or channel mismatch from being mistaken for latent defect.

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1. Status, Scope, and Revision Policy

Primal Calculus studies agreement. The basic setting is a latent comparison domain Λ together with sector reconstructions of the same comparison data into a common additive carrier Y . The reconstructions are not assumed to coincide. They are compared by a cyclic residual. The zero set of that residual is the exact agreement locus.

latent comparison data + fixed chart phase + sector maps $\longrightarrow$ Aleph defect.

Agreement is not primitive. Agreement is measured. Exact realspace is not assumed in advance; it is the residual-zero locus.

Definition 1.1 (Working description). *Primal Calculus is a formal method for measuring whether multiple reconstructions of the same latent object agree in a common equality carrier, while keeping latent agreement, projected image behavior, normal defect sensitivity, and terminal validation distinct.*

This paper is bounded by six rules. Definitions are internal to the formalism. Standard external theorems are cited where invoked. Worked models are calibrations or diagnostics, not empirical discoveries. Broad problem frames are proof-obligation grammars, not replacements for native proof. The cyclic triple is treated as the minimal cyclic model, not as a universal necessity. Provisional branch machinery is separated from the main conceptual spine.

2. Native Mathematical Substrate and Actual Contribution

The algebraic core of this paper lies near standard equalizers, cyclic coboundaries, compatibility conditions, descent data, obstruction language, residual equations, regular level sets, and pullback metrics. This is intentional. Primal Calculus does not claim that equalizers, coboundaries, the constant-rank theorem, or pullback metrics are new.

The contribution is architectural: the calculus fixes a reusable typed interface for reconstruction problems. It asks a modeler to specify a latent comparison domain, sector maps into a common carrier, a residual that measures disagreement, an exact locus, a projection from that locus, and the admissibility and terminal predicates required for an application. The result is a validation grammar that prevents notation from being confused with proof.

substrate: equalizers, coboundaries, regular level sets, pullbacks,

interface: $\Lambda, R_i, \delta_A, \mathcal{A}, M_0, \pi, g_N, g_T, \text{Adm}, Q$.

The calculus is useful when disagreement, projection, and validation would otherwise be conflated. In particular, it separates three questions: whether sector reconstructions agree in the latent comparison domain; whether projection of the exact locus is immersed, injective, or regular; and whether the projected object satisfies an external terminal predicate.

3. Primitive Data

A fixed-phase Primal model is built from typed data.

Definition 3.1 (Fixed-phase Primal model). *A fixed-phase model is a tuple*

$$\mathcal{M} = (\Lambda, A, Y, \theta_0, R_{\theta_0}^\beta, \Xi, \pi_{\theta_0}^\beta),$$

with optional auxiliary data $(B, \beta, \Gamma, J_i, h, \text{Adm})$ when branch geometry, target metrics, or admissibility constraints are needed.

Here Λ is the latent comparison domain; A is the finite cyclic axis set, usually $A = \mathbb{Z}_3$ in the minimal model; Y is the common additive equality carrier; θ_0 is a fixed chart phase; $R_{\theta_0}^\beta$ is a sector reconstruction tuple; Ξ is a defect contrast on the zero-sum defect space; and $\pi_{\theta_0}^\beta$ is the projection defined on the exact locus.

The latent domain is model data. It is not created merely by declaring a symbol. A raw comparison datum may be written schematically as

$$C = (a, b, \delta, \chi),$$

where a and b are raw states, δ is a contrast quantity, and χ is contextual data. Given a transformation class $\mathcal{T}$ and tolerance $\varepsilon_{\text{stab}}$, define the stabilized comparison set by

$$\text{Stab}_{\varepsilon_{\text{stab}}}(S, \mathcal{T}) = \{C : |\delta(a, b; \chi) - \delta(Ta, Tb; T\chi)| \leq \varepsilon_{\text{stab}} \text{ for all } T \in \mathcal{T}\}.$$

A generation or completion rule G and an equivalence relation $\sim$ then determine

$$\Lambda = G(\text{Stab}_{\varepsilon_{\text{stab}}}(S, \mathcal{T})) / \sim.$$

This quotient is a modeling decision, not a theorem. If G , $\sim$, or the topology is unspecified, then Λ remains a placeholder rather than a fully defined mathematical object.

A reconstruction field must descend through the quotient. If $q : X \rightarrow X/\sim$ is the quotient map, a field $f : X \rightarrow Y$ descends only when

$$x \sim x' \implies f(x) = f(x').$$

The same condition applies to every sector map.

4. Why the Cyclic Triple

The main text uses $A = \mathbb{Z}_3$ because three sectors form the minimal non-binary cyclic comparison grammar. A single sector cannot measure agreement. Two sectors measure discrepancy, but the comparison is only bilateral. Three sectors are the first cyclic system in which successor comparison closes into a zero-sum residual circulation.

This is not a claim that nature, mathematics, or every application requires three sectors. The construction generalizes to $\mathbb{Z}_n$. The role of $\mathbb{Z}_3$ is methodological: it is the smallest cyclic model that distinguishes agreement from pairwise opposition while preserving closed residual accounting.

$$\mathbb{Z}_3 : R_0 \rightarrow R_1 \rightarrow R_2 \rightarrow R_0, \quad \mathbb{Z}_n : R_i \rightarrow R_{i+1} \quad (i \bmod n).$$

For applications requiring more than three independent reconstruction channels, the n -sector extension should be used. The triple remains the minimal core calculus because it is the first cyclic case with non-binary disagreement geometry.

5. Cyclic Axis Algebra

Set $A = \mathbb{Z}_3 = \{0, 1, 2\}$, with indices read modulo three. For a triple $R = (R_0, R_1, R_2) \in Y^A$, define the cyclic shift P by

$$(PR)_i = R_{i+1}.$$

Then $P^3 = I$.

Definition 5.1 (Cyclic coboundary). *The cyclic coboundary is*

$$\delta_A = I - P,$$

so that

$$\delta_A(R_0, R_1, R_2) = (R_0 - R_1, R_1 - R_2, R_2 - R_0).$$

Define the diagonal agreement subspace and zero-sum defect subspace by

$$\Delta_Y = \{(y, y, y) : y \in Y\}, \quad Z_Y = \{(W_0, W_1, W_2) \in Y^A : W_0 + W_1 + W_2 = 0\}.$$

Proposition 5.1 (Kernel and image). *For any additive carrier Y ,*

$$\ker \delta_A = \Delta_Y, \quad \text{im } \delta_A = Z_Y.$$

Proof. If $\delta_A R = 0$, then $R_0 = R_1$, $R_1 = R_2$, and $R_2 = R_0$; hence R is diagonal. Conversely, every diagonal triple is annihilated by δ_A . Every cyclic difference triple sums to zero, so $\text{im } \delta_A \subseteq Z_Y$. Conversely, if $W_0 + W_1 + W_2 = 0$, take $R_0 = 0$, $R_1 = -W_0$, and $R_2 = -(W_0 + W_1)$. Then $\delta_A R = W$. $\square$

The diagonal projection is

$$\Pi_\Delta = \frac{1}{3}(I + P + P^2), \quad Q = I - \Pi_\Delta.$$

If the inner product on Y^A makes P unitary, then

$$\delta_A^* = I - P^2, \quad \Delta_A = \delta_A^* \delta_A = 2I - P - P^2 = 3Q.$$

Thus the tests $\delta_A R = 0$ and $QR = 0$ are equivalent.



Figure 1: Basic fixed-phase workflow. The residual is downstream from sector reconstruction; exact agreement is the zero-residual condition.

6. The Aleph Operator

Definition 6.1 (Aleph Operator). *Given a fixed-phase reconstruction tuple*

$$R_{\theta_0}^\beta : \Lambda \rightarrow Y^A, \quad R_{\theta_0}^\beta = (R_{0,\theta_0}^\beta, R_{1,\theta_0}^\beta, R_{2,\theta_0}^\beta),$$

the Aleph Operator is

$$\mathcal{A}_{\theta_0}^\beta = \delta_A \circ R_{\theta_0}^\beta : \Lambda \rightarrow Z_Y.$$

In coordinates,

$$\mathcal{A}_{\theta_0}^\beta(\lambda) = (R_0(\lambda) - R_1(\lambda), R_1(\lambda) - R_2(\lambda), R_2(\lambda) - R_0(\lambda)).$$

The Aleph Operator is not an additional physical assumption. It is the typed cyclic residual determined by the reconstruction tuple.

Definition 6.2 (Exact and approximate loci). *The exact agreement locus is*

$$M_{0,\theta_0}^\beta = (\mathcal{A}_{\theta_0}^\beta)^{-1}(0) \subseteq \Lambda.$$

Given a defect contrast $\Xi : Z_Y \times Z_Y \rightarrow \mathbb{R}$, the finite-tolerance locus is

$$M_{\varepsilon,\theta_0}^\beta = \{\lambda \in \Lambda : \|\mathcal{A}_{\theta_0}^\beta(\lambda)\|_\Xi \leq \varepsilon\}.$$

Proposition 6.1 (Equalizer identity). *For every $\lambda \in \Lambda$,*

$$\mathcal{A}_{\theta_0}^\beta(\lambda) = 0 \iff R_{0,\theta_0}^\beta(\lambda) = R_{1,\theta_0}^\beta(\lambda) = R_{2,\theta_0}^\beta(\lambda).$$

Proof. The residual vanishes exactly when all three cyclic pairwise differences vanish. That is equivalent to equality of the three sector values. $\square$

On the exact locus, the three sectors define one value.

Definition 6.3 (Projection on the exact locus). *Define*

$$\pi_{\theta_0}^\beta : M_{0,\theta_0}^\beta \rightarrow Y, \quad \pi_{\theta_0}^\beta(\lambda) = R_{i,\theta_0}^\beta(\lambda), \quad i \in A.$$

This is well-defined by the equalizer identity. The projected image is

$$U_{\text{real},\theta_0}^\beta = \pi_{\theta_0}^\beta(M_{0,\theta_0}^\beta) \subseteq Y.$$

The exact locus lives in the latent comparison domain. The projected image lives in the carrier. A singularity of the image need not be a singularity of the latent equalizer; it may be created by non-injectivity, non-immersion, quotienting, or projection degeneracy.

7. Fixed-Phase Policy

Earlier live-phase syntax used maps of the form

$$\hat{R}^\beta : \Lambda \times \Theta \rightarrow Y^A$$

while also writing projection as a function of λ alone. That is ill typed unless phase independence has been proved. The corrected core fixes a chart phase $\theta = \theta_0$. With phase fixed, each sector has type

$$R_{i,\theta_0}^\beta : \Lambda \rightarrow Y, \quad i \in A.$$

A live-phase extension is possible, but it must be typed as

$$\widehat{R}^\beta : \Lambda \times \Theta \rightarrow Y^A, \quad \widehat{\mathcal{A}}^\beta = \delta_A \circ \widehat{R}^\beta, \quad \widehat{M}_0^\beta = (\widehat{\mathcal{A}}^\beta)^{-1}(0) \subseteq \Lambda \times \Theta.$$

One must then integrate over $\Lambda \times \Theta$, average over Θ , or choose a section $\theta = s(\lambda)$. The foundational paper keeps the fixed-phase calculus as the main line.

8. Differential Structure and Regularity

Assume now that Λ is a smooth manifold near the point under study, that Y is a finite-dimensional vector space or smooth carrier, and that the sector maps are differentiable on a fixed-branch region. Then

$$D\mathcal{A}_{\theta_0}^\beta = \delta_A \circ DR_{\theta_0}^\beta.$$

Definition 8.1 (Regular exact point). *A point $\lambda \in M_{0,\theta_0}^\beta$ is a regular exact point if $D\mathcal{A}_{\theta_0}^\beta$ has constant rank in a neighborhood of λ .*

Theorem 8.1 (Regular equalizer theorem). *Suppose Λ is smooth near λ , the relevant branch data are fixed on a neighborhood, and $D\mathcal{A}_{\theta_0}^\beta$ has constant rank r near $\lambda \in M_{0,\theta_0}^\beta$. Then M_{0,θ_0}^β is locally a smooth submanifold of codimension r , and*

$$T_\lambda M_{0,\theta_0}^\beta = \ker(D\mathcal{A}_{\theta_0}^\beta|_\lambda).$$

Proof. Apply the constant-rank theorem to the map $\mathcal{A}_{\theta_0}^\beta : \Lambda \rightarrow Z_Y$. The tangent identity is the differential form of the equation $\mathcal{A}_{\theta_0}^\beta = 0$. $\square$

This theorem is local. It does not prove global nonemptiness, connectedness, compactness, or stability under perturbation.

9. Normal Defect Form, Tangential Metric, and Gauge

Two metric constructions appear naturally, and they must not be confused.

Definition 9.1 (Normal defect form). *If Ξ is a defect contrast on Z_Y , define*

$$g_{N,\theta_0}^\beta = (D\mathcal{A}_{\theta_0}^\beta)^*\Xi(D\mathcal{A}_{\theta_0}^\beta),$$

that is,

$$g_{N,\theta_0}^\beta(u, v) = \Xi(D\mathcal{A}_{\theta_0}^\beta(u), D\mathcal{A}_{\theta_0}^\beta(v)).$$

This form measures first-order stiffness of the residual. If $u \in T_\lambda M_{0,\theta_0}^\beta$, then $D\mathcal{A}_{\theta_0}^\beta(u) = 0$, so the form vanishes against every vector. Its degeneracy along exact tangent directions is expected.

Definition 9.2 (Tangential image metric). *If h is a bilinear form on Y , define the tangential image metric on the exact locus by pullback:*

$$g_{T,\theta_0}^\beta = (\pi_{\theta_0}^\beta)^* h, \quad g_{T,\theta_0}^\beta(u, v) = h(D\pi_{\theta_0}^\beta(u), D\pi_{\theta_0}^\beta(v)).$$

The normal form measures sensitivity of agreement. The tangential metric measures geometry induced on the projected image.

Definition 9.3 (Gauge directions). *At $\lambda \in M_{0,\theta_0}^\beta$, a gauge direction is a tangent direction invisible both to the residual and to projection:*

$$G_\lambda \subseteq \ker(D\mathcal{A}_{\theta_0}^\beta|_\lambda) \cap \ker(D\pi_{\theta_0}^\beta|_\lambda).$$

When such directions are present, one may form

$$T_\lambda^{\text{phys}} M = T_\lambda M_{0,\theta_0}^\beta / G_\lambda.$$

Gauge quotienting is not automatic. One must prove that the quotient is well defined and that the desired metric or predicate descends.

10. Reconstruction Dictionary

Once the primitive data are fixed, the reusable dictionary is

$$\begin{aligned} \mathcal{A} &= \delta_A \circ R, & M_0 &= \mathcal{A}^{-1}(0), & M_\varepsilon &= \{\lambda : \|\mathcal{A}(\lambda)\|_\Xi \leq \varepsilon\}, \\ \pi(\lambda) &= R_i(\lambda) \text{ on } M_0, & D\mathcal{A} &= \delta_A \circ DR, & g_N &= (D\mathcal{A})^* \Xi (D\mathcal{A}), & g_T &= \pi^* h. \end{aligned}$$

This dictionary packages definitions and immediate consequences. Nontrivial theorem content enters only when one proves existence, regularity, stability, curvature, admissible minimization, quotient descent, or singularity separation under explicit hypotheses.

11. Worked Example: Three-Channel Reconstruction Audit

This example shows the value of the calculus without claiming new physics or new algebra. Let $\Lambda = \mathbb{R}$, $Y = \mathbb{R}$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the target signal. Three reconstruction channels are

$$R_0(\lambda) = f(\lambda), \quad R_1(\lambda) = f(\lambda) + e_1(\lambda), \quad R_2(\lambda) = f(\lambda) + e_2(\lambda).$$

The Aleph residual is

$$\mathcal{A}(\lambda) = (-e_1(\lambda), e_1(\lambda) - e_2(\lambda), e_2(\lambda)).$$

Therefore

$$\mathcal{A}(\lambda) = 0 \iff e_1(\lambda) = e_2(\lambda) = 0.$$

For the Euclidean defect contrast, the finite-tolerance locus is

$$M_\varepsilon = \left\{ \lambda : e_1(\lambda)^2 + (e_1(\lambda) - e_2(\lambda))^2 + e_2(\lambda)^2 \leq \varepsilon^2 \right\}.$$

The point is not the computation itself. The point is the audit trail. The calculus distinguishes the raw channel mismatch from the exact agreement locus and from whatever projected interpretation is attached later. It therefore blocks the invalid inference “the projected object looks coherent, so the reconstructions agree” and also blocks the reverse invalid inference “the projection is singular, so the latent agreement locus is singular.”

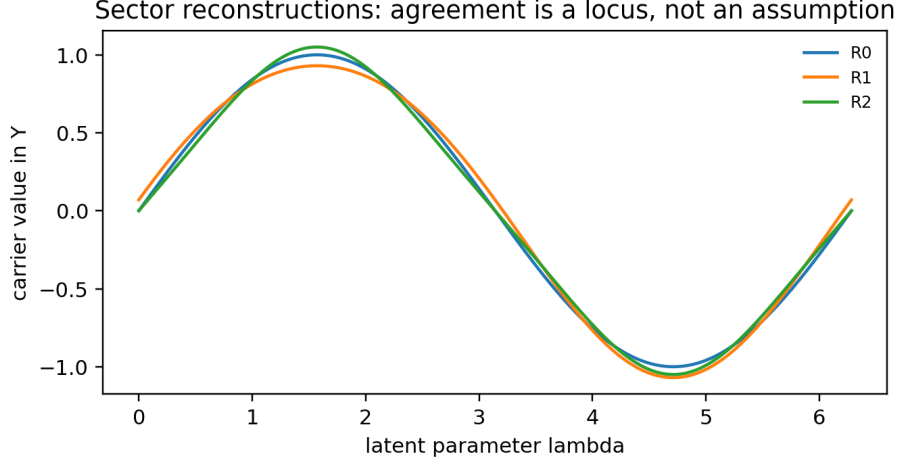


Figure 2: Three sector reconstructions. The calculus treats agreement as a residual-zero condition, not as an assumption.

12. Calibration I: Euclidean Recovery

Let $\Lambda = \mathbb{R}^n$, $Y = \mathbb{R}^n$, and choose a smooth map $f : \Lambda \rightarrow Y$. Define

$$R_0 = f, \quad R_1 = f, \quad R_2 = f.$$

Then $\mathcal{A} = 0$ for all λ , $M_0 = \Lambda$, and $\pi = f$. This is the trivial calibration: if all sectors are identical, the residual vanishes.

A less trivial flat calibration lets each sector be a noisy or transformed channel:

$$R_i(\lambda) = f(\lambda) + e_i(\lambda), \quad i = 0, 1, 2.$$

Then

$$\mathcal{A}(\lambda) = (e_0 - e_1, e_1 - e_2, e_2 - e_0).$$

The exact locus is precisely the set where $e_0 = e_1 = e_2$. The finite-tolerance locus is the region where channel mismatch is bounded by ε .

13. Calibration II: Hyperbolic Mass-Shell Reconstruction

This calibration records how a curved image may arise from an exact latent reconstruction. Let the carrier include a split-signature plane with coordinates (E, p) and target relation

$$E^2 - p^2 = m^2.$$

For $m > 0$, introduce rapidity ψ and define

$$\Gamma_H(m, \psi, \sigma) = \sigma m (\cosh \psi, \sinh \psi), \quad \sigma \in \{+1, -1\}.$$

Then

$$E^2 - p^2 = m^2 (\cosh^2 \psi - \sinh^2 \psi) = m^2.$$

A cyclic model can place three sector reconstructions into a common carrier and impose equality through $\mathcal{A} = 0$. The hyperbolic branch supplies projected image geometry; the Aleph residual supplies agreement among sectors. The point is not to derive relativity. The point is to show that a non-Euclidean target relation can be represented as the projection of exact cyclic agreement once the branch map is specified.

14. Calibration III: Projection Cusp

The cusp example separates latent regularity from image singularity. Let $\Lambda = \mathbb{R}$ with coordinate t , and let $Y = \mathbb{R}^2$. Define identical sector maps

$$R_0(t) = R_1(t) = R_2(t) = (t^2, t^3).$$

Then $\mathcal{A}(t) = 0$ for all t , so $M_0 = \mathbb{R}$ is smooth. The projection is

$$\pi(t) = (t^2, t^3).$$

The image satisfies

$$y^2 = x^3,$$

which has a cusp at the origin. The latent equalizer is regular, while the projected image is singular because $D\pi(0) = 0$.

latent agreement $\neq$ projection immersion $\neq$ target image regularity.

A singular target image does not automatically imply a singular latent agreement locus.

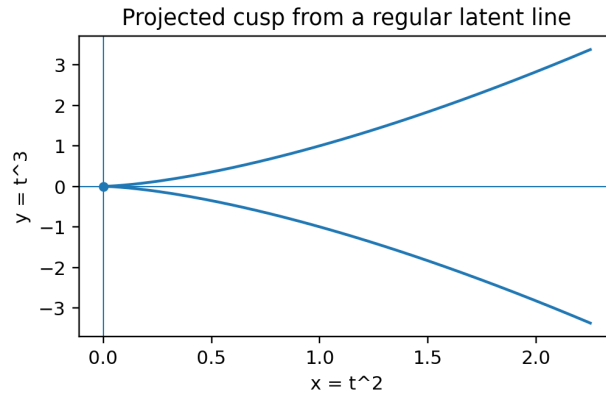


Figure 3: A regular latent line projects to a cusp. The singularity belongs to the projection image, not to the latent equalizer.

15. Variational Use

A residual calculus naturally admits variational expressions. Given a measure μ on Λ and defect contrast Ξ , define the fixed-phase Aleph action

$$S[R] = \int_{\Lambda} \|\mathcal{A}(\lambda)\|_{\Xi}^2 d\mu(\lambda).$$

This action measures total sector disagreement. Minimizers, if they exist, are best-agreement reconstructions under the chosen admissible class. Exact minimizers with $S = 0$ lie in the zero-residual regime.

However, the action does not guarantee existence of minimizers. A serious variational theorem must specify coercivity, lower semicontinuity, compactness, admissible convergence, and closure of the terminal predicate. Approximate kernels are useful only if one proves a limit-stability theorem connecting ε -agreement to exact native agreement.

16. External Problem Frames and Certificate Grammar

For a broader problem P , a fixed-phase Primal problem frame has the form

$$P_{\theta_0} = (\Lambda_P, Y_P, R_{P,0}, R_{P,1}, R_{P,2}, \text{Adm}_P, \pi_P, T_P, Q_P).$$

Here Λ_P is the comparison domain, Y_P is the problem-specific equality carrier, $\text{Adm}_P \subseteq \Lambda_P$ is the native admissible class, T_P is the target class, and Q_P is the terminal predicate. The problem Aleph Operator and exact admissible sector are

$$\mathcal{A}_P = \delta_A(R_{P,0}, R_{P,1}, R_{P,2}), \quad M_{P,0} = \text{Adm}_P \cap \mathcal{A}_P^{-1}(0).$$

This is a formal equalizer representation. It becomes a serious problem translation only when the native mathematical data and the cyclic data are equivalent on admissible classes.

Let N_P denote the native objects, hypotheses, proof data, and equivalence relations of problem P . Let C_P denote the cyclic data satisfying the frame syntax. A faithful correspondence consists of maps

$$\Phi_P : N_P \rightarrow C_P, \quad \Psi_P : C_P \rightarrow N_P,$$

such that

$$\Psi_P \circ \Phi_P = \text{id}_{N_P}, \quad \Phi_P \circ \Psi_P = \text{id}_{C_P}$$

on admissible equivalence classes, and such that

$$\mathcal{A}_P(\Phi_P(N)) = 0 \iff N \text{ satisfies the native equations and compatibility laws,}$$

$$Q_P(\Phi_P(N)) \iff N \text{ satisfies the native terminal statement.}$$

The forced certificate forms are:

- **Section:** $\sigma : T_P \rightarrow \text{Adm}_P$ with $\pi_P \sigma = \text{id}_{T_P}$ and $\mathcal{A}_P \sigma = 0$.
- **Separator:** an invariant $I : \text{Adm}_P \rightarrow W$ separating exact admissible data from a bad sector.
- **Witness:** $\lambda^* \in \text{Adm}_P$ with $\mathcal{A}_P(\lambda^*) = 0$ and a positive terminal margin.

Naming the certificate form is not the same as constructing the certificate. Construction must occur in the native mathematics.

17. Claim Classification

Claim class	Role in this paper
Definition	Primitive data, cyclic coboundary, Aleph Operator, exact locus, approximate locus, projection, normal defect form, tangential metric, and problem-frame grammar are internal definitions.
Algebraic theorem	Kernel, image, and equalizer identities follow from the cyclic coboundary.
Differential theorem	Local regularity follows from the constant-rank theorem when its hypotheses hold.
Diagnostic example	Euclidean recovery, mass-shell reconstruction, projection cusp, and three-channel audit illustrate the grammar; they do not establish empirical laws.
Certificate grammar	Broad mathematical problems can be encoded only after admissibility, projection, target, and terminal predicate are fixed.
Open theorem target	Branch classification, stability, minimization, gauge descent, and limit passage remain unresolved unless separately proved.

This classification is included to prevent notation from hardening into unsupported conclusion.

18. Known Unknowns and Unknown Unknowns

Object	Known unknowns	Unknown unknowns
Latent domain	How to choose G , $\sim$, topology, and smooth structure in each application.	Quotient pathologies or hidden comparison data excluded by the chosen latent carrier.
Branch palette	Whether $\{E, P, H\}$ follows from a minimal classification principle.	Additional branch families not represented by current syntax.
Aleph kernel	Nonemptiness, regularity, connectedness, compactness, and global structure.	Equalizer components invisible to first-order residual tests.
Projection	Faithfulness, immersion, injectivity, surjectivity, and quotient compatibility.	Projection degeneracies that manufacture false image geometry.
Variational theory	Coercivity, lower semicontinuity, compactness, and admissible convergence.	Limit failures that preserve residual smallness but lose terminal meaning.
Native problem frames	Whether selected carriers and sectors preserve all native hypotheses and equivalences.	Obstruction classes outside the chosen cyclic carrier.
Empirical use	Observation-to-latent and projection-to-measurement maps remain to be specified.	Measurement artifacts that imitate exact agreement.

19. Validity Criteria

A serious application of Primal Calculus must satisfy the following criteria.

1. Typed primitives: Λ , Y , A , $R_{\theta_0}^\beta$, Ξ , and π must be specified with domains and codomains.
2. Descent: reconstruction fields must descend through the latent quotient.
3. Carrier adequacy: Y must preserve the quantities required by the target problem.
4. Residual correctness: $\mathcal{A} = 0$ must be equivalent to the intended compatibility laws.
5. Projection faithfulness: π must preserve terminal meaning.
6. Admissibility: the admissible class must not smuggle in the desired conclusion.
7. Regularity hypotheses: differential claims require smoothness, fixed branch, and rank conditions.
8. Limit stability: approximate or regularized data must converge without loss of terminal predicate.
9. Native certificate: for external problems, a section, separator, or witness must be constructed in native mathematics.

The ease of writing formal symbols makes self-deception easy. A framework designed to organize difficult statements needs a discipline for distinguishing syntactic organization from faithful translation.

20. Current Theorem Targets

The next theorem targets are branch classification, atlas construction for multiple fixed-phase charts, cusp-resolution generalization, existence of minimizers for Aleph actions, perturbative stability, gauge quotient descent, and limit-stability for certificates. These targets are not promises. They mark the frontier between the stable core and the theorem program still required for a deeper theory.

21. Positioning Statement

Primal Calculus is a formal reconstruction calculus in which multiple sector maps from a latent comparison domain are compared in a common equality carrier. The Aleph Operator is the cyclic residual measuring disagreement. Its zero set is the exact equalizer. Its finite sublevel sets are approximate neighborhoods of agreement. Its differential controls local regularity and normal defect stiffness. Its projection map determines the geometry of the resulting image.

This statement is intentionally narrower than a universal-foundation claim and broader than a single worked example. The framework is a typed grammar for organizing agreement, defect, projection, and validation. It is not an empirical theory and not an automatic proof engine.

22. Conclusion

Primal Calculus begins with a tuple of sectorwise reconstructions from a latent comparison domain into a common additive carrier. The cyclic coboundary turns that tuple into the Aleph residual. The exact locus is the zero set of that residual. Projection of that exact locus into the carrier yields an image that may be flat, curved, singular, or degenerate.

The main conceptual distinction is separation of roles. Latent agreement is one object. Projection behavior is another. Normal defect geometry is distinct from tangential image geometry. These distinctions are elementary once stated, but they become operationally useful when a projected image is singular while the latent equalizer remains regular.

A. Provisional Branch Machinery

The provisional branch palette used in geometric reconstructions is

$$B = \{E, P, H\}.$$

The meanings are: E for elliptic fixed-radius circular branch; P for polar variable-radius circular branch; and H for hyperbolic variable-radius split-signature branch. For circular branches,

$$\Gamma_E(\rho, \psi) = \rho(\cos \psi, \sin \psi), \quad \Gamma_P(\rho, \psi) = \rho(\cos \psi, \sin \psi),$$

with the distinction that the elliptic branch freezes the radius while the polar branch reads a variable latent radius. For the hyperbolic branch,

$$\Gamma_H(\rho, \psi, \sigma) = \sigma \rho(\cosh \psi, \sinh \psi), \quad \sigma \in \{+1, -1\}.$$

These satisfy

$$u^2 + v^2 = \rho^2 \quad (E, P), \quad u^2 - v^2 = \rho^2 \quad (H).$$

In geometric cases with $Y = \mathbb{R}^3$ and basis (e_0, e_1, e_2) , define insertion maps

$$J_i : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad J_i(u, v) = ue_{i+1} + ve_{i+2},$$

with indices modulo three. A sector reconstruction may be written

$$R_{i, \theta_0}^\beta(\lambda) = a_i(\lambda)e_i + J_i \Gamma_{\beta_i(\lambda)}(\rho_i(\lambda), \theta_0 + \alpha_i(\lambda), \sigma_i(\lambda)).$$

The branch palette remains provisional until derived from a classification theorem.

B. Rederivation Checklist

The shortest stable reconstruction path is: start with three maps $R_0, R_1, R_2 : \Lambda \rightarrow Y$; set $A = \mathbb{Z}_3$ and $R = (R_0, R_1, R_2) \in Y^A$; define $\delta_A(R) = (R_0 - R_1, R_1 - R_2, R_2 - R_0)$; prove $\ker \delta_A = \Delta_Y$ and $\text{im } \delta_A = Z_Y$; define $\mathcal{A} = \delta_A R$; define $M_0 = \mathcal{A}^{-1}(0)$; prove $\mathcal{A} = 0$ iff $R_0 = R_1 = R_2$; define $\pi(\lambda) = R_i(\lambda)$ on M_0 ; differentiate $DA = \delta_A DR$; apply the constant-rank theorem; define $g_N = (DA)^* \Xi(DA)$ and $g_T = \pi^* h$; then diagnose whether singular behavior belongs to the latent equalizer, the projection, or the image.

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